**P P Savani University**
**School of Engineering**
**Department of Information Technology**

Course Code: SEIT3270
Course Name: Web Technologies
Prerequisite Course(s): --

**Teaching & Examination Scheme:**

| Teaching Scheme (Hours/Week) | | | | Examination Scheme (Marks) | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Theory | Practical | Tutorial | Credit | Theory | | Practical | | Tutorial | | Total |
| | | | | CE | ESE | CE | ESE | CE | ESE | |
| 3 | 2 | 0 | 5 | 40 | 60 | 40 | 60 | 0 | 0 | 200 |

CE: Continuous Evaluation, ESE: End Semester Exam

**Objective(s) of the Course:**
To help learners to
- understand the basic components of a computer system.
- identify an appropriate approach to computational problems.
- develop logic building and problem-solving skills.

**Course Content:**

| **Section I** | | | |
|---|---|---|---|
| Module No. | Content | Hours | Weightage in % |
| 1. | **Introduction to Web Design:** Concept of WWW, Internet and WWW, HTTP Protocol : Request and Response, Web browser and Web servers, Concepts of effective web design, Web design issues including Browser, Display resolution, Page Layout and linking, User centric design, Sitemap, Planning and publishing website, Designing effective navigation | 07 | 20 |
| 2. | **HTML:** Basics of HTML, formatting and fonts, commenting code, color, hyperlink, lists, tables, images, forms, XHTML, Meta tags, Character entities, frames and frame sets, Browser architecture and Web site structure. Overview and features of HTML5 | 08 | 20 |
| 3. | **Style sheets:** Need for CSS, introduction to CSS, basic syntax and structure, using CSS, background images, colors and properties, manipulating texts, using fonts, borders and boxes, margins, padding lists, positioning using CSS, CSS2, Overview and features of CSS3 | 07 | 10 |
| **Section II** | | | |
| Module No. | Content | Hours | Weightage in % |
| 4. | **JavaScript:** Client side scripting with JavaScript, variables, functions, conditions, loops and repetition, Pop up boxes, Advance JavaScript: Javascript and objects, JavaScript own objects, the DOM and web browser environments, Manipulation using DOM, forms and validations, DHTML : Combining HTML, CSS and Javascript, Events and buttons. | 08 | 20 |
| 5. | **PHP:** Introduction and basic syntax of PHP, decision and looping with examples, PHP and HTML, Arrays, Functions, Browser control and detection, string, Form processing, Files, Advance Features: Cookies | 08 | 20 |

| | | | |
|---|---|---|---|
| | and Sessions, Object Oriented Programming with PHP | | |
| 6. | **MySQL:**<br>Basic commands with PHP examples, Connection to server, PHP my admin and database bugs | 07 | 10 |
| | **TOTAL** | 45 | 100 |

**List of Practical:**

| Sr. No. | Name of Practical | Hours |
|---|---|---|
| 1. | Design web a page for the department timetable containing a suitable header, footer, description of the course codes and titles, faculty name and initialises, etc., use href, list tags and add University logo as a background. Using table tag and additional features like spanning rows, columns and table borders. [Refer your class timetable] | 02 |
| 2. | a) Create a form to collect Student feedback. (Use textbox, text area, checkbox, radio button, select box etc.).<br>b) Create a web page using frame. Divide the page into two parts with Navigation links on left hand side of page (width=20%) and content page on right hand side of page (width = 80%). On clicking the navigation Links corresponding content must be shown on the right-hand side. | 03 |
| 3. | Create a web page using frame. Divide the page into two parts with Navigation links on left hand side of page (width=20%) and content page on right hand side of page (width = 80%). On clicking the navigation Links corresponding content must be shown on the right-hand side. | 02 |
| 4. | Write HTML code to develop a webpage having two frames that divide the webpage into two equal rows and then divide the row into equal columns fill each frame with a different background colour. | 03 |
| 5. | Create your resume using HTML tags also experiment with colors, text (bold, italic and different headings), image, link, size and also other related tags. | 02 |
| 6. | Design a web page using CSS with suitable design for the following:<br>i.Demonstrate different font styles<br>ii.Control the repetition of image with background-repeat property<br>iii.Define style for links as a: link, a: active, a: hover, a: visited<br>iv. Demonstrate Element visibility property | 03 |
| 7. | Design a web page of your institute with an attractive background color, text color, an Image, font etc. (use internal CSS). Use External CSS to format the class timetable. | 02 |
| 8. | Develop simple calculator for Addition, Subtraction, Multiplication and Division operation using JavaScript.<br>a) Create HTML Page that contains form with fields Name, Email, Mobile No, Gender, Favorite Color and a button. Write a JavaScript code to validate all the fields when the button is clicked, later combine and display the information in textbox. | 03 |
| 9. | Write an XHTML document which displays a form containing text elements to input register number, sub-code, marks in three tests and a button element. Also write Java script compute the average of two best tests on click of button and print average marks using alert. Validate all the fields using JavaScript. | 03 |
| 10. | a) Write a PHP script to display today's date in dd-mm-yyyy format.<br>b) Write a PHP script to check whether the number is prime or not when user input a valid number from client side. | 03 |
| 11. | Create HTML page that contain textbox, submit / reset button. Write PHP script to display this information and also store into a text file. | 02 |
| 12. | Write a PHP script for login authentication. Design an HTML form which takes Username and Password from user and validate against stored Username and Password in a file. | 02 |
| | **TOTAL** | 30 |

**Text Book(s):**

| Title | Author/s | Publication |
|---|---|---|
| Web Design with HTML, CSS, JavaScript and jQuery | by Jon Duckett | Wiley-India |
| | | |

**Reference Book(s):**

| Title | Author(s) | Publication |
|---|---|---|
| Web Technologies: HTML, JAVASCRIPT, PHP, JAVA, JSP, XML and AJAX, Black Book | Kogent Learning Solutions Inc. | Dreamtech Press |
| Developing Web Applications | Ralph Moseley and M. T. Savaliya | Wiley-India |

**Web Material Link(s):**

- Web Technology PDF Download | WT Books, Lecture Notes, Studymaterial

**Course Evaluation:**

**Theory:**

- Continuous Evaluation consists of two tests, each of 30 marks and 1 hour of duration and average of the same will be converted to 30 marks.
- Faculty evaluation consists of 10 marks as per the guidelines provided by the Course Coordinator.
- End Semester Examination consists of 60 marks.

**Practical:**

- Continuous Evaluation consists of practical performance which should be evaluated out of 10 for each practical and average of the same will be converted to 20 marks.
- Internal viva consists of 20 marks.
- Practical performance/quiz/test consists of 30 marks during End Semester Exam.
- Viva-voce consists of 30 marks during End Semester Exam.

**Course Outcome(s):**

After completion of the course, the students will be able to

| SEIT3270 | WEB TECHNOLOGIES |
|---|---|
| CO 1 | Understand the fundamental concepts of web technologies |
| CO 2 | Understand and implement client-side and server-side programming |
| CO 3 | Learn HTML, CSS, and JavaScript for web page development |
| CO 4 | Work with databases for dynamic web applications |

**Mapping of CO with PO**

| SEIT3270 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | P010 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO 1 | 1 | | | | | | | | | | | |
| CO 2 | 1 | | 1 | | | | | | | | | |
| CO 3 | | | | | 1 | | | | | | | |
| CO 4 | | | | 1 | | | | | | | | |
| CO 5 | | | | | | | 1 | | | | | |

**Mapping of CO with PSO**

| SEIT3270 | PSO1 | PSO2 | PSO3 |
|---|---|---|---|
| CO 1 | 1 | | |

| | | | |
|---|---|---|---|
| CO 2 | | 1 | |
| CO 3 | 1 | | |
| CO 4 | 1 | | |
| CO 5 | | 1 | |

**Level of Revised Bloom's Taxonomy in Assessment**

| | | |
|---|---|---|
| 1: Remember | 2: Understand | 3: Apply |
| 4: Analyze | 5: Evaluate | 6: Create |

| Module No | Content | RBT Level |
|---|---|---|
| 1. | Introduction to Web Design | 1,2 |
| 2. | HTML | 1,2,3 |
| 3. | Style sheets | 3,4 |
| 4. | JavaScript | 2,3,4 |
| 5. | PHP | 2,3 |
| 6. | MySQL | 2,3 |